

AYMANE AIT DADS

Research Intern - Data Science & ML | Sensor Fusion · Forecasting · Operational Dashboard

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PROFESSIONAL SUMMARY

Built and deployed machine-learning pipelines on 100,000+ telecom network records at Orange Maroc, reducing manual analysis time by ~60% and delivering actionable KPIs directly to a network optimization team via Power BI dashboards. Ranked 17th out of 3,000+ teams (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge, demonstrating end-to-end model training, ablation methodology, and diagnostic evaluation at scale. Skilled in Python, Pandas, feature engineering, clustering, ensemble methods, and interactive data visualization - tools directly applicable to building congestion forecasting models and operational decision-support frameworks. Aims to contribute to Fujitsu's Space Data Frontier Research Center by developing ML-driven vessel ETA forecasting and port-state KPI dashboards for the San Pedro Bay port complex.

CORE COMPETENCIES

decision-support framework | congestion forecasting | machine-learning models | sensor fusion | operational dashboard | feature engineering | ETA forecasting | multi-source data integration

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built Random Forest and K-Means models on **100,000+ fiber-optic network records** (upload speed, latency, jitter) to classify network quality across thousands of nodes, directly supporting operational decision-making.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with pipeline output adopted directly by the network optimization team for infrastructure planning.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation workflows, accelerating the team's ability to diagnose network congestion and prioritize maintenance actions.
- Delivered stakeholder presentations translating model outputs into **actionable maintenance recommendations**, using Power BI dashboards to communicate KPIs and congestion patterns to non-technical decision-makers.

PROJECTS

Aerial Cactus Detection - Endangered Species Classification

EURECOM · 2025 · 99.8% Accuracy

- Benchmarked CNN, DenseNet121, and hybrid CNN-Transformer architectures on **17,500+ drone images**, achieving 99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0 for endangered species classification from aerial imagery.
- Designed end-to-end image classification pipeline with **multi-model benchmarking** and ensemble evaluation, directly applicable to SAR/optical imagery analysis for vessel detection and port-area monitoring.

PyTorch · DenseNet121 · CNN · Transformer · Kaggle

NVIDIA Nemotron Model Reasoning Challenge

Kaggle · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built diagnostic evaluation pipeline revealing that over-expanded reasoning - not incorrect reasoning - was the primary failure mode.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, Cloud Computing, NLP, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Data Science & Analytics: Python | SQL | Pandas | NumPy | Scikit-learn | Power BI | feature engineering

Machine Learning & Modeling: PyTorch | Random Forest | K-Means clustering | ensemble methods | CNN | DenseNet | LoRA SFT

MLOps & Visualization: Git | Linux | FP16/BF16 mixed precision | Matplotlib | Seaborn | Docker (basic) | ablation studies